Passenger satisfaction and airline service
Micro-Project 2

https://github.com/ChalesManyang/ANA500-Micro-Projects

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November 7, 2026

Problem Statement

Which passenger and service characteristics are associated with airline satisfaction?

The airline needs evidence to identify service areas for further investigation. This observational dataset supports associations, but cannot establish the effects of service changes.

Hypothesis Formulation

Hypothesis: higher online boarding and comfort ratings are associated with greater satisfaction.

Satisfaction may also differ by travel class.

The visual analysis assesses descriptive patterns. Observational associations cannot establish that changing a service will cause higher satisfaction.

Acquire

Source: user-supplied airline.csv, read with pandas.read_csv.

129,880 passenger records and 25 original columns. The file contains demographics, travel class, service ratings, delays and a satisfaction category.

No collection dates, airline names or sampling documentation accompany the file. Row order does not establish a time sequence.

Prepare

Preserve the raw DataFrame and clean an independent copy.

Remove the exported row counter and trim categorical text. Retain passenger ID only for integrity checks. No duplicate rows or repeated IDs occur.

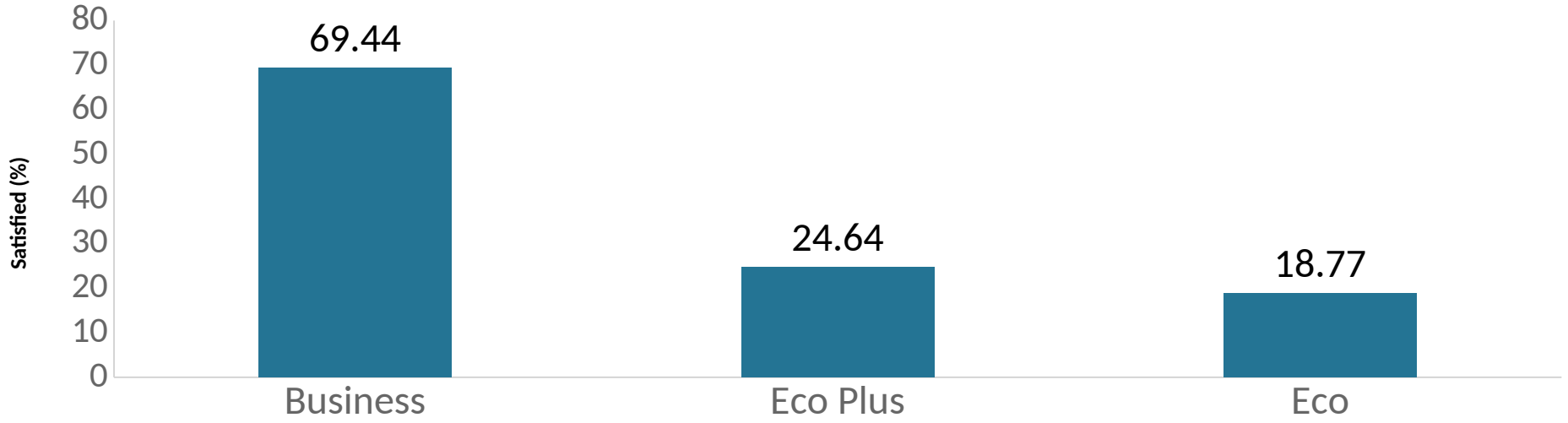
Arrival delay has 393 missing values (0.30%). Preserve these as unknown. Keep valid ratings from 0 to 5 because zero meaning is undocumented.

NumPy: known arrival-delay mean 15.09 minutes and median 0. Age and nonnegative delay checks pass.

Analyze data

Descriptive analysis uses all available passenger records. Satisfaction equals 1 for satisfied and 0 otherwise.

Satisfied passengers: 43.45% overall. Class differences may also reflect travel purpose and passenger composition.



Report

Online boarding has the strongest service-rating association with satisfaction (r = 0.502).

Arrival delays have a median of 0 and a long right tail. The notebook includes delay histograms and a departure-arrival scatter plot.

Priorities for investigation are descriptive. Class and customer differences may confound these patterns. Project 2 ends at Report.

